

Supersymmetry breaking & the BPS BEH effect:

* Susy is Broken when at least one aux. field does not vanish at the minimum of the potential.

* Since the theory is Gauged we expect the Gauged fields to obtain mass; Gravitino

* The Potential of the theory, V , has $V_+ + V_-$
so if the mass terms from $\bar{W}$ and f_{AB} make
 $V > 0$: Cosmological const. $\Lambda > 0$ dS
 $V < 0$: No mass terms: $\Lambda < 0$ AdS

Susy breaking & the Goldstone Fermion:

Toy model: Take 1-chiral multiplets. $(Z^A, P_L \chi^A, F^A)$

w/ Potential $W(Z)$

$$S := \int d^4x \left[-\partial^\mu \bar{Z}_a \partial_\mu Z^a - \bar{\chi}_a \not{\partial} P_L \chi^a - \bar{W}^a W_a - \frac{1}{2} \chi^a P_L W_{ab} \chi^b - \frac{1}{2} \bar{\chi}_a P_R \bar{W}^{ab} \chi_b \right]$$

Minimize $V(Z, \bar{Z}) = W_a(Z) \bar{W}^a(Z)$

$Z^a = Z^a_0$, stationary pts s.t.

$$\partial_a V = W_{ab} \bar{W}^b|_{Z_0} = 0$$

* Here W_{ab} has zero eigenvectors $\bar{W}^B$. We know $\bar{W}^B \neq 0$ since $V > 0$. Here W_{ab} is the ferm. mass matrix.

$$\mathcal{L} = -\bar{\psi}_i \not{\partial} P_L \psi^i + \frac{1}{2} M_{ij} \bar{\psi}^i P_L \psi^j + \frac{1}{2} \bar{M}^{ij} \bar{\psi}_i P_R \psi_j$$

$$\text{E.O.M: } \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu P_L \psi^i} \right) = \frac{\partial \mathcal{L}}{\partial \psi^i}$$

$$\not{\partial} P_L \psi^i = \bar{M}^{ij} P_R \psi_j \quad | \quad \not{\partial} P_R \psi_i = M_{ij} P_L \psi^j$$

Suppose M_{ij} has a zero eigenvalue $\bar{V}^i$:

$$M_{ij} \bar{V}^j = 0$$

$$P_L V \equiv V_i P_L \psi^i$$

is the massless ferm. To see this note:

$$\not{D} P_L \psi = V_i \not{D} P_L \psi^i = V_i \bar{m}^{ij} P_R \psi_j = 0$$

We can apply this to our n -chiral mltfs $(\chi^\alpha, \bar{\psi}^\alpha, F^\alpha)$

Idea: Fermion term is

$$(e \text{ vector or } U_{AB}) \chi P_L \chi^\alpha \leftarrow \text{(term of theory)}$$

$$\therefore W_{\alpha\beta} \bar{W}^\beta = 0 \quad \text{mut. by } -\frac{1}{\sqrt{2}} \text{ to have}$$

$$U_\alpha = -\frac{1}{\sqrt{2}} W_\alpha, \quad \text{so}$$

$$P_L V = -\frac{1}{\sqrt{2}} W_\alpha P_L \chi^\alpha = \not{D}_L \chi^\alpha (\not{D}_S P_R \chi_\alpha)$$

$$\boxed{\not{D}_S P_L V = \not{D}_S P_L \chi^\alpha g_{\alpha\bar{\beta}} \not{D}_S P_R \chi^{\bar{\beta}}}$$

* For a Gauged $N=2$ Susy action: $V = g^{\alpha\bar{\beta}} W_\alpha \bar{W}_{\bar{\beta}} + \frac{1}{2} \text{Re}(F^A) P_A P_B$

$$\partial_\alpha V = m_{\alpha\bar{\beta}} \bar{W}^{\bar{\beta}} + m_{\alpha A} (-i P^A / \sqrt{2}) = 0$$

This is the "top row" of

$$\begin{pmatrix} m_{\alpha\bar{\beta}} & m_{\alpha A} \\ m_{\bar{\alpha}\beta} & m_{\bar{\alpha} A} \end{pmatrix} \begin{pmatrix} \bar{W}^{\bar{\beta}} \\ -i P^B / \sqrt{2} \end{pmatrix} = 0$$

$$\text{Here } -\sqrt{2} \begin{pmatrix} \not{D}_S P_L \chi^{\bar{\beta}} \\ \not{D}_S P_L \lambda^B \end{pmatrix} = \bar{V}_i$$

$$\therefore P_L V = V_i P_L \sum_\alpha (\text{fermions})^\alpha i \\ = -\frac{1}{\sqrt{2}} (W_\alpha P_L \chi^\alpha + \frac{i}{\sqrt{2}} P_A \lambda^A)$$

$$\boxed{P_L V = -\frac{1}{\sqrt{2}} P_L (W_\alpha \chi^\alpha + \frac{i}{\sqrt{2}} P_A \lambda^A)}$$

$$P_L V = P_L (\chi^\alpha g_{\alpha\bar{\beta}} \not{D}_S P_R \chi^{\bar{\beta}} + \lambda^A (P_C(F) / \sqrt{2})_{AB} \not{D}_S P_R \lambda^B)$$

Returning to the Sup. conf. model:

$$\mathcal{L}_{\text{mix}} = -(\bar{F} \cdot \gamma) P_L \psi + \text{h.c.}$$

$$P_L \psi = -\frac{1}{\sqrt{2}} \chi^\alpha e^{iK/2} \nabla_\alpha \bar{W} - \frac{1}{2} i P_L \lambda^A \bar{P}_A$$

Here we have used $P_L \psi = -\frac{1}{\sqrt{2}} P_L (W_I \Omega^\pm) + \dots$

$$-\frac{1}{\sqrt{2}} P_L (W_I - \chi^\alpha \gamma \nabla_\alpha \bar{Z}^F)$$

$$\text{where } \gamma W_I \nabla_\alpha \bar{Z}^F = \nabla_\alpha W = \gamma^3 \nabla_\alpha \bar{W}$$

$$\Rightarrow -\frac{1}{\sqrt{2}} P_L (\gamma^3 \nabla_\alpha \bar{W} \chi^\alpha) = -\frac{1}{\sqrt{2}} \nabla_\alpha \bar{W} e^{iK/2} \chi^\alpha$$

which is where the first term comes from.

We can express this term in terms of fermion shifts:

$$P_L \psi = \chi^\alpha \delta_\psi \chi_\alpha + P_L \lambda^A \delta_\psi P_R \lambda_A$$

where we previously found

$$\delta_\psi \chi_\alpha = g_{\alpha\bar{\beta}} \delta_\psi \chi^{\bar{\beta}} = -\frac{1}{\sqrt{2}} e^{iK/2} \nabla_\alpha \bar{W}$$

$$\delta_\psi P_R \lambda_A = \text{Re}(f_{AB}) \delta_\psi P_R \lambda^B = -\frac{1}{2} i P_A$$

Recall the super potential had negative contributions from the gravitino + contribution from other fields.

$$V_- = -3K^2 e^{iK/2} W \bar{W}$$

$$V_+ = e^{iK/2} \nabla_\alpha W g^{\alpha\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} + \frac{1}{2} (\text{Re}(f)^{-1})^{AB} P_A P_B$$

By varying the Goldstino ψ under susy:

$$\delta_Q (P_L \psi) = -\frac{1}{\sqrt{2}} \delta_Q \chi^\alpha e^{iK/2} \nabla_\alpha \bar{W}$$

$$-\frac{1}{2} i \delta_Q (P_L \lambda^A) \bar{P}_A$$

$$= -\frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} \right) \left(\delta \chi^\alpha - e^{iK/2} g^{\alpha\bar{\beta}} \bar{\nabla}_{\bar{\beta}} \bar{W} \right) e^{iK/2} \nabla_\alpha \bar{W}$$

$$-\frac{1}{2} i \left[\frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^A + \frac{1}{2} i \text{Re}(f)^{AB} P_B \right] \epsilon P_A$$

$$= \frac{1}{2} (e^{K^2 K} g^{\alpha\bar{\beta}} \bar{\psi}_{\bar{\beta}} \not{W} \psi_{\alpha} W + \frac{1}{2} (\text{Re}(f))^{-1} Y^{AB} P_A P_B) P_L \epsilon$$

$$= \frac{1}{2} V + P_L \epsilon + \dots$$

$$\underline{S_0(P_L V)} = \underline{\frac{1}{2} V + P_L \epsilon} + \dots$$

Since $V_f > 0$ in a Susy-breaking vacuum, the Goldstino undergoes a shift under susy. From this we can impose the following susy gauge condition:

$$\boxed{W = 0} \quad (\text{Susy gauge condition})$$

which eliminates the goldstino from the theory.

[Recall a global symm is broken if $T_A|0\rangle \neq 0$. At the quantum level; Susy is broken if $Q_A|0\rangle \neq 0$.

The above Gauge choice eliminates mixing of $\bar{\psi}_m$ w/ other fermions.

$$\mathcal{L}_{mix} = (\bar{\psi} \cdot \gamma) \left[\frac{1}{2i} P_A \not{A} P_A + \frac{1}{2} \gamma^{\alpha} e^{K^2 K/2} \not{V}_{\alpha} W \right] + \text{h.c.}$$

$$\mathcal{L}_m = \frac{1}{2} K^2 e^{K^2 K/2} W \bar{\psi}_m P_R \gamma^{\mu\nu} \psi_{\mu}$$

$$\rightarrow \mathcal{L}_{mix} = -(\bar{\psi} \cdot \gamma) P_L V + \text{h.c.}$$

Thus, the Goldstino mixes these terms. and so

here $\mathcal{L}_{mix} = 0$ and we are left w/

$$\mathcal{L}_m = \frac{1}{2} K^2 e^{K^2 K/2} W \bar{\psi}_m P_R \gamma^{\mu\nu} \psi_{\mu} + \dots$$

We interpret this term as a Higgs mass

$$|m_{3/2}|^2 \equiv \frac{1}{2} K^2 e^{K^2 K/2} W$$

this is the Super BHE effect. of the Rarita-Schwinger action.

* When scalars are constant and other terms in the action are neglected, the gravitino terms become

$$e^{-1} \mathcal{L} = \frac{1}{2\kappa^2} \bar{\Psi}_m (-\gamma^{\mu\nu\rho} \partial_\nu \Psi_\rho + (m_{3/2} P_R + \bar{m}_{3/2} P_L) \gamma^{\mu\rho}) \Psi_\rho - \bar{\Psi}_m \gamma^\mu \mathcal{N}$$

We now attempt to diagonalize this Lagrangian by introducing the massive gravitino fields:

$$P_L \Psi_m = P_L \Phi_m - \frac{2\kappa^2}{3|m_{3/2}|^2} \partial_m P_L \mathcal{N} - \left(\kappa^2/3\bar{m}_{3/2}\right) \gamma_m P_L \mathcal{N}$$

This corresponds to a susy transformation w/ the appropriate parameters:

$$\delta_Q P_L \Phi_m = (\partial_m P_L \epsilon) + \frac{1}{2} \kappa^2 \gamma_m e^{\kappa^2 K/2} \omega P_R \epsilon$$

$$\text{w/ } m_{3/2} = \kappa^2 e^{\kappa^2 K/2} \omega$$

$$\delta_Q P_L \Phi_m = (\partial_m P_L \epsilon) + \frac{1}{2} \gamma_m m_{3/2} P_R \epsilon$$

$$\begin{aligned} \therefore \delta_Q P \Phi_m &= \partial_m P_L \epsilon + \frac{1}{2} \gamma_m m_{3/2} P_R \epsilon \\ &\quad - \frac{2\kappa^2}{3|m_{3/2}|^2} \partial_m \left(\frac{1}{2} V_\mu P_L \epsilon \right) + \left(\frac{\kappa^2}{3\bar{m}_{3/2}} \gamma_m \frac{1}{2} V_\mu P_R \epsilon \right) \\ &= \partial_m P_L \epsilon \left(1 - \frac{\kappa^2}{3m} \right) + \gamma_m \frac{\kappa^2}{3\bar{m}_{3/2}} P_R \epsilon \dots \end{aligned}$$

So we simply have different values for the Q-susy.

$$e^{-1} \mathcal{L} = \frac{1}{2\kappa} \bar{\Psi}_m (-\gamma^{\mu\nu\rho} \partial_\nu + (m_{3/2} P_R + \bar{m}_{3/2} P_L) \gamma^{\mu\rho}) \Psi_\rho + \frac{\kappa^2}{3|m_{3/2}|^2} (\bar{\mathcal{N}} \partial \mathcal{N} + 2 \bar{\mathcal{N}} (m_{3/2} P_R + \bar{m}_{3/2} P_L) \mathcal{N})$$

$$\begin{aligned} \text{Proof: } & (\bar{\Psi}_m - \alpha \partial_m \bar{\mathcal{N}} - \beta \partial_m \bar{\mathcal{N}})(m) (\Psi_m - \alpha \partial_m \mathcal{N} - \beta \partial_m \mathcal{N}) \\ &= \bar{\Psi}_m \Psi_m + (-\alpha \partial_m \bar{\mathcal{N}})(m) (-\alpha \partial_m \mathcal{N}) + (-\alpha \partial_m \bar{\mathcal{N}})(m) (-\beta \partial_m \mathcal{N}) \\ &\quad + (-\beta \partial_m \bar{\mathcal{N}})(m) (-\alpha \partial_m \mathcal{N}) + (-\beta \partial_m \bar{\mathcal{N}})(m) (-\beta \partial_m \mathcal{N}) \end{aligned}$$

- $-\alpha \partial_m \bar{\psi} (-\gamma^{m\rho} \partial_\rho \psi \leftarrow \text{symm } (m_{(m\rho)} \text{ anti-}^{(mn)})$
- $(\alpha \partial_m \bar{\psi} (m_{3/2} P_R + \bar{m}_{3/2} P_L) \gamma^{m\rho} (-\beta \gamma_{\rho} \psi))$
 $= -\alpha \beta \partial_m \bar{\psi} (m_{3/2} P_R + \bar{m}_{3/2} P_L) 3 \gamma^\rho \psi$
 $= -\alpha \beta \cancel{\partial} \bar{\psi} (m_{3/2} P_L + \bar{m}_{3/2} P_R) \psi (3)$
 $+ \underline{\hspace{1cm}}$
 $= \partial_m (\alpha \beta \bar{\psi} (m) \psi)$
 This term vanishes as $\partial M^4 = 0$

$\rightarrow \beta^2 \gamma_{\mu} \bar{\psi} (-\gamma^{m\rho} \partial_\rho \psi + (m_{3/2} P_R + \bar{m}_{3/2} P_L) \gamma^{m\rho}) \gamma_\mu \psi$
 $\beta^2 \gamma_{\mu} \bar{\psi} (- (2) \gamma^{m\rho} \partial_\rho \psi + (m) 3 \gamma^m) \psi$
 $\bar{\psi} \frac{\beta^2}{2\kappa^2} (-6 \cancel{\partial} + m 12) \psi$
 $= \frac{\kappa^2}{4 \bar{m}_{3/2}} (\bar{\psi}) (-3 \cancel{\partial} + m 6) \psi$
 $\frac{\kappa^2}{3 \bar{m}_{3/2}} (\bar{\psi}) (-\cancel{\partial} + \underline{2m}) \psi$

• Missing a - sign to cancel last term:

$-\bar{\psi}_m(m) \alpha \partial_\rho \psi - \beta \bar{\psi}_m(m) \gamma_\rho \psi$
 $= -\bar{\psi}_m (-\gamma^{m\rho} \partial_\rho \psi + (m_{3/2} P_R + \bar{m}_{3/2} P_L) \gamma^{m\rho}) \partial_\rho \psi$
 $= -\bar{\psi}_m (m_{3/2} P_R + \bar{m}_{3/2} P_L) \gamma^{m\rho} \partial_\rho \psi \propto$
 $-\beta \bar{\psi}_m (-\gamma^{m\rho} \partial_\rho \psi + m \gamma^{m\rho}) \gamma_\rho \psi$
 $= -\bar{\psi}_m(m) \gamma^{m\rho} \partial_\rho \psi \propto -\beta \bar{\psi}_m (-2 \gamma^{m\rho} \partial_\rho \psi + m 3 \gamma^m) \psi$
 $\underbrace{\hspace{10em}}_{\text{terms cancel}}$

- $(-\alpha \partial_m \bar{\psi} - \beta \cancel{\partial} \bar{\psi}) (\gamma^{m\rho} \partial_\rho \psi + m \gamma^{m\rho}) \psi_\rho$
 $-\alpha (\partial_m \bar{\psi}) m \gamma^{m\rho} \psi_\rho + \beta \cancel{\partial} \bar{\psi} \gamma^{m\rho} \partial_\rho \psi \psi_\rho - \beta \bar{\psi} 3 m \cancel{\psi}$

$$\sim \partial_m (\quad)$$

→ These terms:

$$= \frac{1}{m_{3/2}^2} \bar{\psi}_m (m_{3/2} P_R + \bar{m}_{3/2} P_L) \gamma^m$$

Not sure where the other term comes from.

* Consider a model w/ one chiral mult. and no gauge fields

Claim: The kinetic term of the Goldstino χ cancels the kinetic term of the chiral fermion if $V=0$

Proof:

The kinetic term of χ is given by

$$e^{-1} \mathcal{L}_{\chi, \text{kin}} = g_{\alpha\beta} \left(\frac{1}{2} \bar{\chi}^\alpha \not{D}^{(0)} \chi^\beta + \frac{1}{2} \bar{\chi}^\beta \not{D}^{(0)} \chi^\alpha \right)$$

Here we simply take the "pure" kinetic term

$$= g_{\alpha\beta} \left(\frac{1}{2} \bar{\chi}^\alpha \gamma^m \partial_m \chi^\beta + \frac{1}{2} \bar{\chi}^\beta \gamma^m \partial_m \chi^\alpha \right)$$

∴ The combined Lagrangian elements is:

$$e^{-1} \mathcal{L}_{\text{un}} = -\frac{1}{2} \left(g_{\alpha\beta} \bar{\chi}^\alpha \gamma^m \partial_m \chi^\beta + \frac{K^2}{3|m_{3/2}|^2} \bar{W} \not{D} W \right)$$

define $C \equiv K^2/3m^2$, $m \equiv m_{3/2}$

Since $\bar{W}$ Gauge multiplets, this implies no

K_A terms $\Rightarrow P_A = 0$

$$\therefore W = -\frac{1}{\sqrt{2}} \chi^\alpha e^{K^2 K/2} \nabla_\alpha W$$

Define $S_\alpha \equiv -\frac{1}{\sqrt{2}} e^{K^2 K/2} \nabla_\alpha W$

$$\therefore W = S_\alpha \chi^\alpha$$

The effective action is given by

$$\underline{G_{\alpha\beta} \equiv g_{\alpha\beta} - C S_\alpha \bar{S}_\beta}$$

Must impose that $G_{\alpha\beta}$ has rank $(n-2)$ and hence
after susy breaking has $(n-1)$ chiral fermions

$$\Rightarrow \det(G) = 0$$

$$G^{\alpha\bar{\beta}_1} \dots G^{\alpha\bar{\beta}_n} \in \epsilon_{\alpha_1 \beta_1} \dots \epsilon_{\alpha_n \beta_n} \dots$$

$$\Rightarrow C = \frac{1}{S_\alpha S_\alpha}$$

$$\Rightarrow S^\alpha S_\alpha = \frac{1}{2} e^{K/2} G^{\alpha\bar{\beta}} \bar{\nabla}_\alpha W \bar{\nabla}_{\bar{\beta}} W = \frac{3}{2} e^K W \bar{W} = \frac{3}{2} m^2 \quad \square$$

* Elimination of the N goldstino term leads to new contributions to the mass matrix:

$$P_L W = \chi^\alpha \delta_\alpha \chi_\alpha + P_L \lambda^A \delta_A P_R \lambda_A$$

which we plug into the $\mathcal{L}^1$ Lag. which we've just derived:

$$\frac{1}{3} \frac{K^2}{m_{3/2}^2} m_{3/2} \bar{\nu} P_R P_R \nu + \dots$$

$$= \frac{2}{3} \frac{K^2}{m_{3/2}} (\delta_\alpha \chi_\alpha \chi^\alpha + \delta_A \lambda^A P_R P_R \lambda^A)$$

we obtain the contributions:

$$m_{\alpha\beta}^{(v)} = -\frac{4K^2}{3m_{3/2}} (\delta_\alpha \chi_\alpha) (\delta_\beta \chi_\beta)$$

$$m_{\alpha A}^{(v)} = -\frac{4K^2}{3m_{3/2}} (\delta_\alpha \chi_\alpha) (\delta_A P_R \lambda_A)$$

$$m_{AB}^{(v)} = -\frac{4K^2}{3m_{3/2}} (\delta_A P_R \lambda_A) (\delta_B P_R \lambda_B)$$

The full mass matrices:

$$m_{\alpha\beta}^g = m_{\alpha\beta} + m_{\alpha\beta}^{(v)} \quad m_{\alpha A}^g = m_{\alpha A} + m_{\alpha A}^{(v)} \quad m_{AB}^g = m_{AB} + m_{AB}^{(v)}$$

* Recall, we had $\partial_\alpha V$ which was proportional to $m_{\alpha\beta}, m_{\alpha A}, m_{AB}$:

$$\partial_\alpha V = e^{K/2} (-2\bar{m}_{3/2} \bar{\nabla}_\alpha W + m_{\alpha\beta} g^{\beta\bar{\gamma}} \bar{\nabla}_{\bar{\gamma}} W) - \frac{i}{\sqrt{2}} P_A (\text{Re}(F))^{\dagger A} m_{\alpha\beta}$$

$$= -\sqrt{2} \delta_\alpha \chi_\beta m_{\alpha\beta} g^{\beta\bar{\gamma}} + 2\sqrt{2} \bar{m}_{3/2} \delta_\alpha \chi_\alpha$$

$$= \sqrt{2} \delta_\alpha \chi_\beta (m_{\alpha\beta} + m_{\alpha\beta}^{(v)})$$

$$-\frac{1}{\sqrt{2}} \partial_\alpha V = \delta_S \chi^\beta m_{\beta\beta} - 2\bar{m}_{3/2} \delta_S \chi_\alpha + \delta_S P_L \lambda^\beta m_{\alpha\beta}$$

* We found Gauge invariance of the superpotential W takes the form:

$$K_A \nabla_\alpha W + \frac{1}{2} \kappa^2 P_A \bar{W} = W_\alpha K_A^\alpha + \kappa^2 \bar{W} \nabla_\alpha W = 0$$

This can be written as:

$$m_{\beta\beta} \delta_S \chi^\beta - 2\bar{m}_{3/2} \delta_S P_L \lambda^\alpha = 0$$

Proof:

$$\left[i\sqrt{2} \left(\frac{1}{2} P_A - \frac{1}{4} f_A \right) (2e^{K/2})^{P_C} P_C \right] \left(-\frac{1}{\sqrt{2}} e^{K/2} \nabla_\alpha W \right) - 2\bar{m}_{3/2} \left(-\frac{1}{2} P_A \right) = 0$$

Putting this together:

$$\text{Now, } m_{3/2} = \kappa^2 e^{K/2} W$$

$$\begin{aligned} & (m_{\alpha\beta}^g - m_{\alpha\beta}^{(v)}) \delta_S \chi^\beta + (m_{\alpha A}^g - m_{\alpha A}^{(v)}) \delta_S P_L \lambda^A \\ & - 2\kappa^2 e^{K/2} \bar{W} \delta_S \chi_\alpha = -\frac{1}{\sqrt{2}} \partial_\alpha V \\ & m_{\alpha\beta}^g \delta_S \chi^\beta + m_{\alpha A}^g \delta_S P_L \lambda^A - 2\kappa^2 e^{K/2} \left(-\frac{1}{\sqrt{2}} e^{K/2} \nabla_\alpha W \right) \bar{W} \\ & = \underline{m_{\alpha\beta}^{(v)} \delta_S \chi^\beta - m_{\alpha A}^{(v)} \delta_S P_L \lambda^A} = -\frac{1}{\sqrt{2}} \partial_\alpha V \end{aligned}$$

$$\begin{aligned} & \rightarrow \frac{4\kappa^2}{3m_{3/2}} (\delta_S \chi_\alpha) (\delta_S \chi_\beta) (\delta_S \chi^\beta) \\ & + \frac{4\kappa^2}{3m_{3/2}} (\delta_S \chi_\alpha) (\delta_S P_L \lambda^A) \delta_S P_L \lambda^A \\ & = \frac{2\kappa^2}{3m_{3/2}} \partial_S \chi_\alpha (2(\delta_S \chi)^2 + 2(\delta_S P_L \lambda)^2) \\ & = C (\delta_S \chi_\alpha) (V + 3\kappa^2 e^{K/2} W \bar{W}) \end{aligned}$$

$$\therefore m_{\alpha\beta}^g + m_{\alpha A}^g + \delta_S \chi_\alpha (C V + C 3\kappa^2 e^{K/2} W \bar{W} - 2\bar{m}_{3/2})$$

$$\begin{aligned} & \rightarrow \left(\frac{2\kappa^2}{3m_{3/2}} \right) 3m_{3/2} \bar{W} e^{K/2} - 2\bar{m}_{3/2} \\ & = 2\bar{m}_{3/2} - 2\bar{m}_{3/2} = 0. \end{aligned}$$

$$m_{\alpha\beta} \delta_S \chi^\beta + m_{\alpha A} \delta_S P_L \chi^A + \frac{2\kappa^2}{3m_{3/2}} V \delta_S \chi_\alpha = -\sqrt{2} \partial_\alpha V$$

* Invariance of the superpotential becomes:

$$m_{\alpha\beta}^g \delta_S \chi^\beta + m_{\alpha 0}^g \delta_S P_L \chi^0 + \frac{2\kappa^2}{3m_{3/2}} V \delta_S P_R \chi_A = 0$$

In a vacuum we have $\partial_\alpha V = 0$. For zero cosmological constant ($\langle V \rangle = 0$) the above conditions show that the full mass matrix has a zero eigenvector

Goldstino in SUGRA:

The fermionic mass matrix of gauged susy has a zero mode. Global susy has these zeros $\rightarrow$ Goldstino, a massless particle.

In SUGRA the zero modes corresponds to the reduction of rank of the full mass matrix m^g because χ (the Goldstino) disappears.

$$\begin{pmatrix} m_{\alpha\beta} & m_{\alpha A} \\ m_{A0} & m_{AB} \end{pmatrix} \begin{pmatrix} \delta_S \chi^\beta \\ \delta_S P_L \chi^B \end{pmatrix} = \begin{pmatrix} -\frac{2\kappa^2}{3m_{3/2}} V \delta_S \chi_\alpha \\ -\frac{2\kappa^2}{3m_{3/2}} V \delta_S P_R \chi_A \end{pmatrix}$$

Extensions to Cosmological Sol^{ns}:

* In cosmological applications of $\mathcal{N}=1$ SUGRA, scalar fields need not be constant. there are then new mixing terms of the gravitino and chiral fermions which involves derivatives of scalars

$$\mathcal{L}_{new} = \frac{1}{\sqrt{2}} g_{\alpha\bar{\beta}} \bar{\Psi}_m \hat{\partial} \bar{z}^{\bar{\beta}} \gamma^m \chi^\alpha + h.c.$$

$$= (\bar{\Psi} \cdot \gamma) \frac{1}{\sqrt{2}} g_{\alpha\bar{\beta}} \hat{\partial} z^\alpha \chi^{\bar{\beta}} + \sqrt{2} g_{\alpha\bar{\beta}} \bar{\Psi}_m \gamma^m \hat{\partial}_n \bar{z}^{\bar{\beta}} \chi^\alpha + h.c.$$

→ Not sure about this expansion note the second has $\bar{z}^{\bar{\beta}}$ and the first z^α .

The first term can be interpreted as the extra condition

$$P_L V' = -\frac{1}{\sqrt{2}} g_{\alpha\bar{\beta}} \hat{\partial} z^\alpha \chi^{\bar{\beta}}$$

to the Goldstino, second term is a mixing

• For cosmology: $N + N'$: Goldstino

$$\delta_Q P_L N = -\frac{1}{2} e^{K/2} \nabla_\alpha U \hat{\partial} z^\alpha P_L \epsilon + \frac{1}{2} V_+ P_L \epsilon$$

$$\delta_Q P_L N' = +\frac{1}{2} e^{K/2} \nabla_\alpha W \hat{\partial} z^\alpha P_L \epsilon - \frac{1}{2} \hat{\partial} z^\alpha g_{\alpha\bar{\beta}} \hat{\partial} \bar{z}^{\bar{\beta}} P_L \epsilon$$

$$\delta_Q P_L (N + N') = \frac{1}{2} (-\hat{\partial} z^\alpha g_{\alpha\bar{\beta}} \hat{\partial} \bar{z}^{\bar{\beta}} + V_+) P_L \epsilon + \dots$$

which is positive for cosmological solutions.